

Operational dashboard

DASHBOARD DESIGN CONCEPTS



Olga Scrivner, PhD

President, Scrivner Solutions Inc

Dashboard taxonomies



Domain: industry type and business metrics

Level: enterprise-wide, departmental, individual

Update frequency: near real-time, monthly, quarterly, annually

Mode

- **Interactivity:** static, interactive
- **Screen:** tablet, desktop, conference monitor

¹ Few, Stephen. (2006). Information dashboard design.

Business roles taxonomy



Management roles

- Junior
- Middle
- Senior

Business activity

- Monitoring daily operations
- Analyzing trends
- Planning long-term strategic goals

Choosing the right dashboard

What problems are you trying to solve?

- Access to operational data or aggregated indicators

¹ Stojanovic, Filip. (2022). Understanding Business Dashboard Types.

Choosing the right dashboard

What problems are you trying to solve?

Who will use the dashboard?

- Access to time-sensitive data, essential indicators, or trends
- Managers and their teams, executives, or analysts

¹ Stojanovic, Filip. (2022). Understanding Business Dashboard Types.

Choosing the right dashboard

What problems are you trying to solve?

Who will use the dashboard?

What is your performance gap?

- Access to time-sensitive data, essential indicators, or trends
- Managers and their teams, executives, or analysts
- Daily or monthly performance and performance issues

¹ Stojanovic, Filip. (2022). Understanding Business Dashboard Types.

Choosing the right dashboard

What problems are you trying to solve?

Who will use the dashboard?

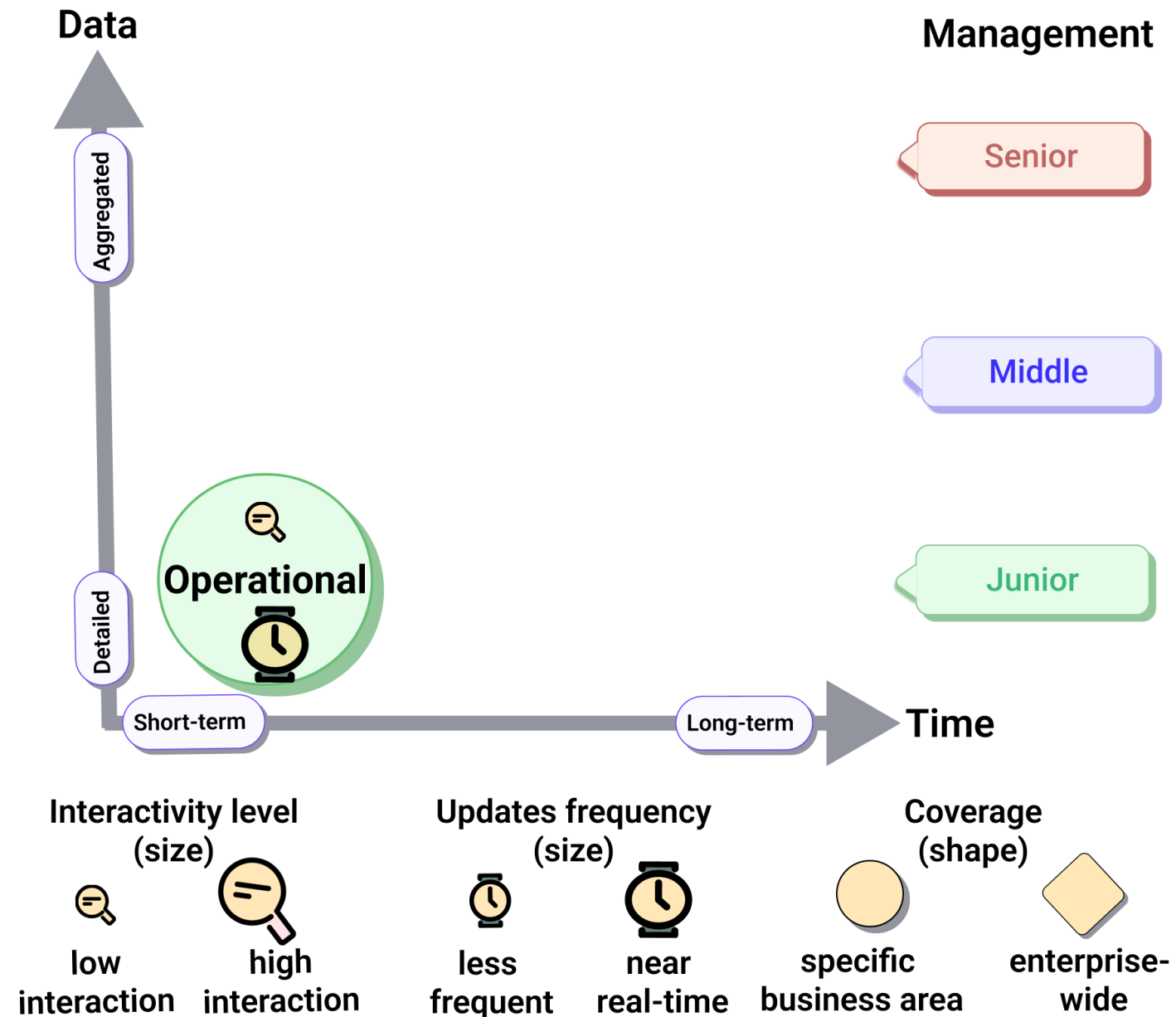
What is your performance gap?

What are your goals?

- Access to time-sensitive data, essential indicators, or trends
- Managers and their teams, executives, or analysts
- Daily or monthly performance and performance issues
- Track performance, achieve targets, set goals

¹ Stojanovic, Filip. (2022). Understanding Business Dashboard Types.

The everyday dashboard



Purpose

- A snapshot of the daily performance

Data Type

- Operational KPIs, short-term data, frequent updates (real-time, daily, weekly)

Interactivity

- Low interaction required

Stakeholders

- Junior management and their teams

Operational dashboard benefits

Predictive analysis

- Seeing the present and predicting the future based on data

¹ FineReport. (2021). Operational Dashboard.

Operational dashboard benefits

Predictive analysis

Active management

- Seeing the present and predicting the future based on data
- Focusing on goals and identifying patterns and trends

¹ FineReport. (2021). Operational Dashboard.

Operational dashboard benefits

Predictive analysis

Active management

Real-time visibility

- Seeing the present and predicting the future based on data
- Focusing on goals and identifying patterns and trends
- Monitoring organizational health through data visualization

¹ FineReport. (2021). Operational Dashboard.

Operational dashboard benefits

Predictive analysis

Active management

Real-time visibility

Continuous improvement

- Seeing the present and predicting the future based on data
- Focusing on goals and identifying patterns and trends
- Monitoring organizational health through data visualization
- Promoting continuous improvement by simplifying work and increasing profitability

¹ FineReport. (2021). Operational Dashboard.

Operational dashboard types

Every business can monitor company's operations and activities



What's currently going on?

Marketing: campaign success, acquisition costs, the number of clicks

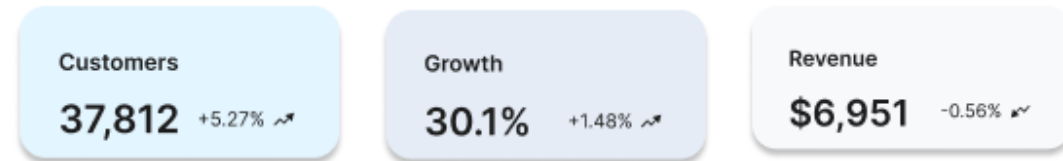
Manufacturing: production volumes, quantity ordered, returned items

Human resources: count of open positions, employee turnover, satisfaction

IT support: resolved cases, number of calls

Operational dashboard graphical components

Status



Status: Indicators (operational KPIs)

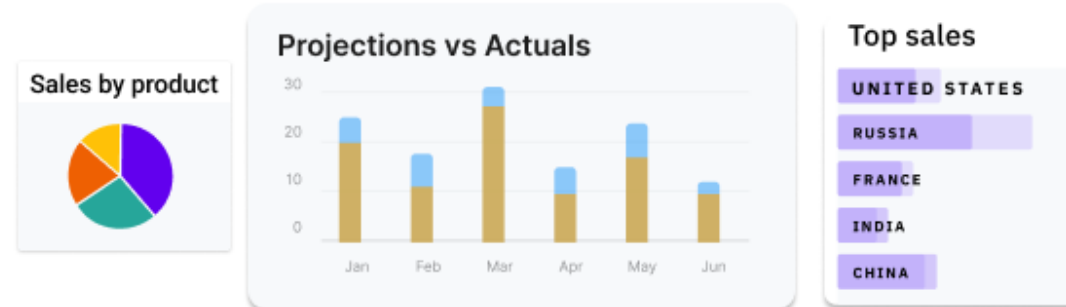
Operational dashboard graphical components

Status



Status: Indicators (operational KPIs)

Comparison



Comparison: bar graphs, bullet charts

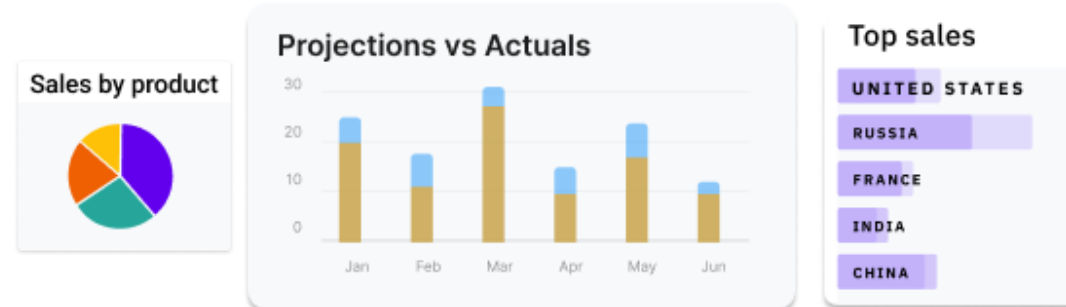
Operational dashboard graphical components

Status



Status: Indicators (operational KPIs)

Comparison



Comparison: bar graphs, bullet charts

Changes



Change: short-term trends

Let's practice!
DASHBOARD DESIGN CONCEPTS

Analytical dashboard

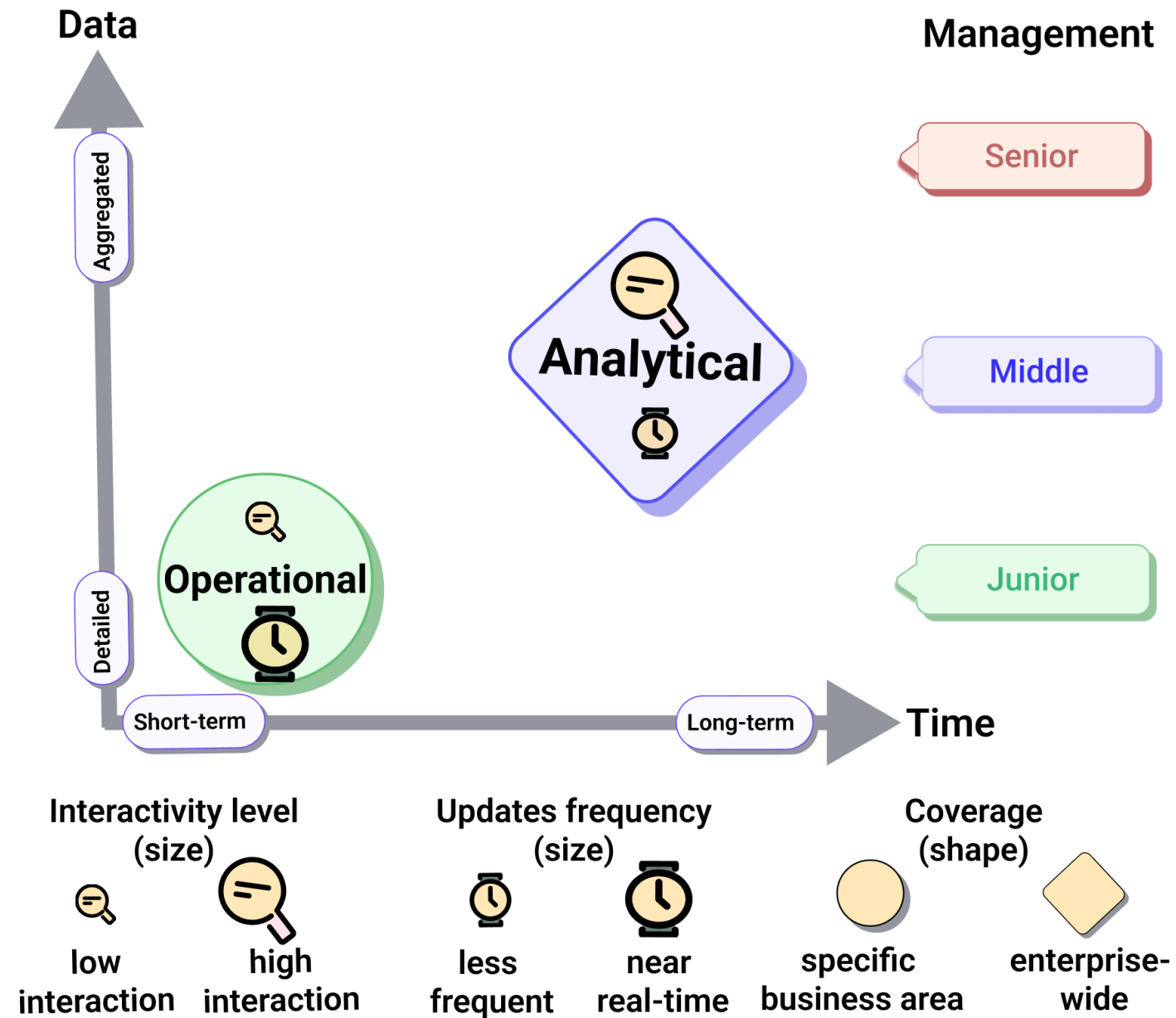
DASHBOARD DESIGN CONCEPTS



Olga Scrivner, PhD

President, Scrivner Solutions Inc

The deep-dive dashboard



Purpose

- Data mining to uncover deeper insights

Data Type

- Large volumes of data, less frequent updates (monthly, quarterly)

Interactivity:

- High interactivity: drill-down, filters, sliders

Stakeholders

- Analysts

Analytical dashboard benefits

Predictive analysis

- Guiding future decisions based on historical data

Analytical dashboard benefits

Predictive analysis

Anomaly detection

- Guiding future decisions based on historical data
- Identifying unusual behavior in business trends

Analytical dashboard benefits

Predictive analysis

Anomaly detection

Pattern discovery

- Guiding future decisions based on historical data
- Identifying unusual behavior in business trends
- Uncovering hidden patterns and how variables interact



Analytical dashboard benefits

Predictive analysis

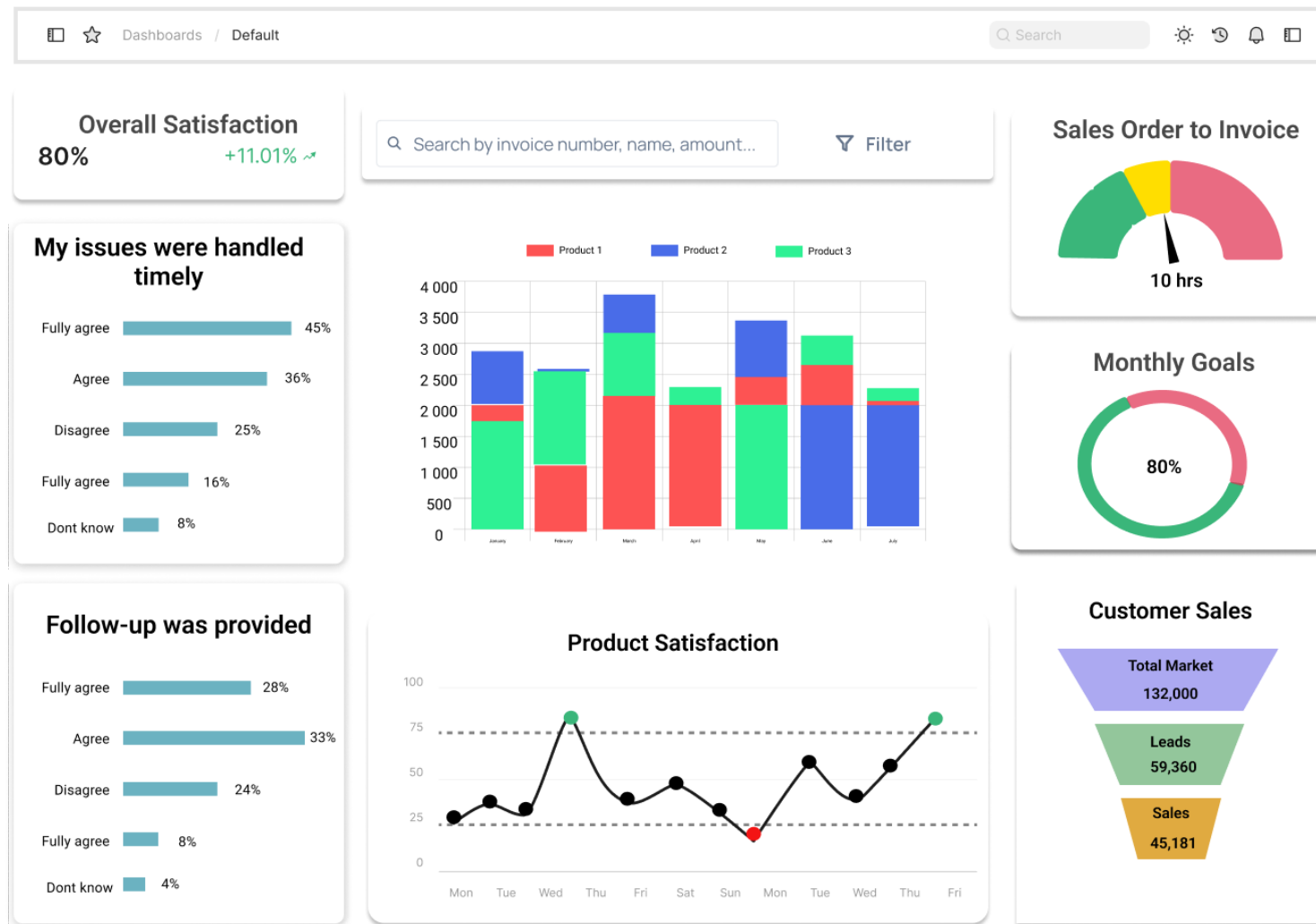
Anomaly detection

Pattern discovery

Trend analysis

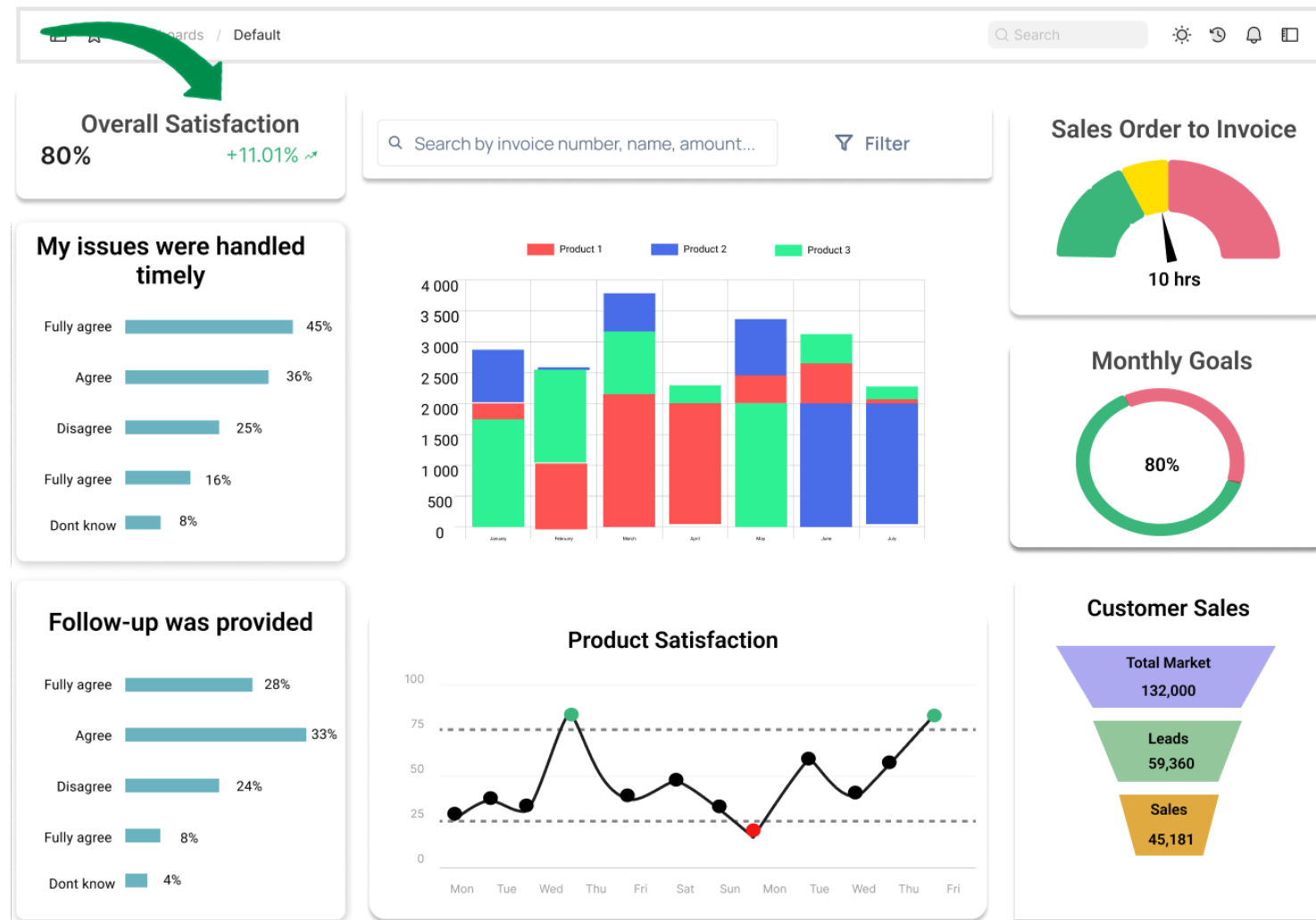
- Guiding future decisions based on historical data
- Identifying unusual behavior in business trends
- Uncovering hidden patterns and understanding how variables interact
- Looking at performance metrics over time to identify problematic periods in the past

Analytical dashboard graphical components



¹ Zoho analytics. 2017. Analytical dashboard.

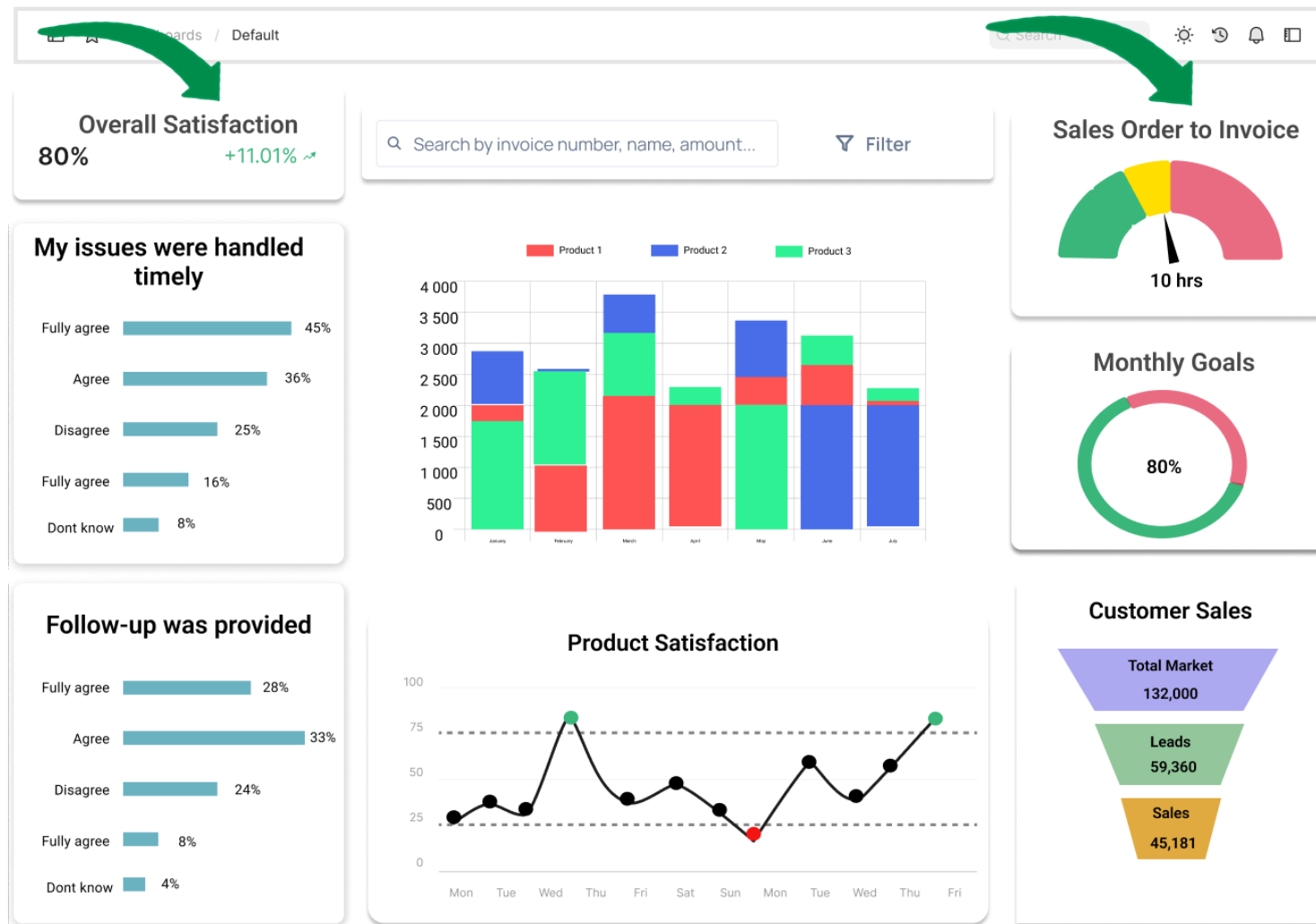
Analytical dashboard graphical components



Status: Indicators

¹ Zoho analytics. 2017. Analytical dashboard.

Analytical dashboard graphical components

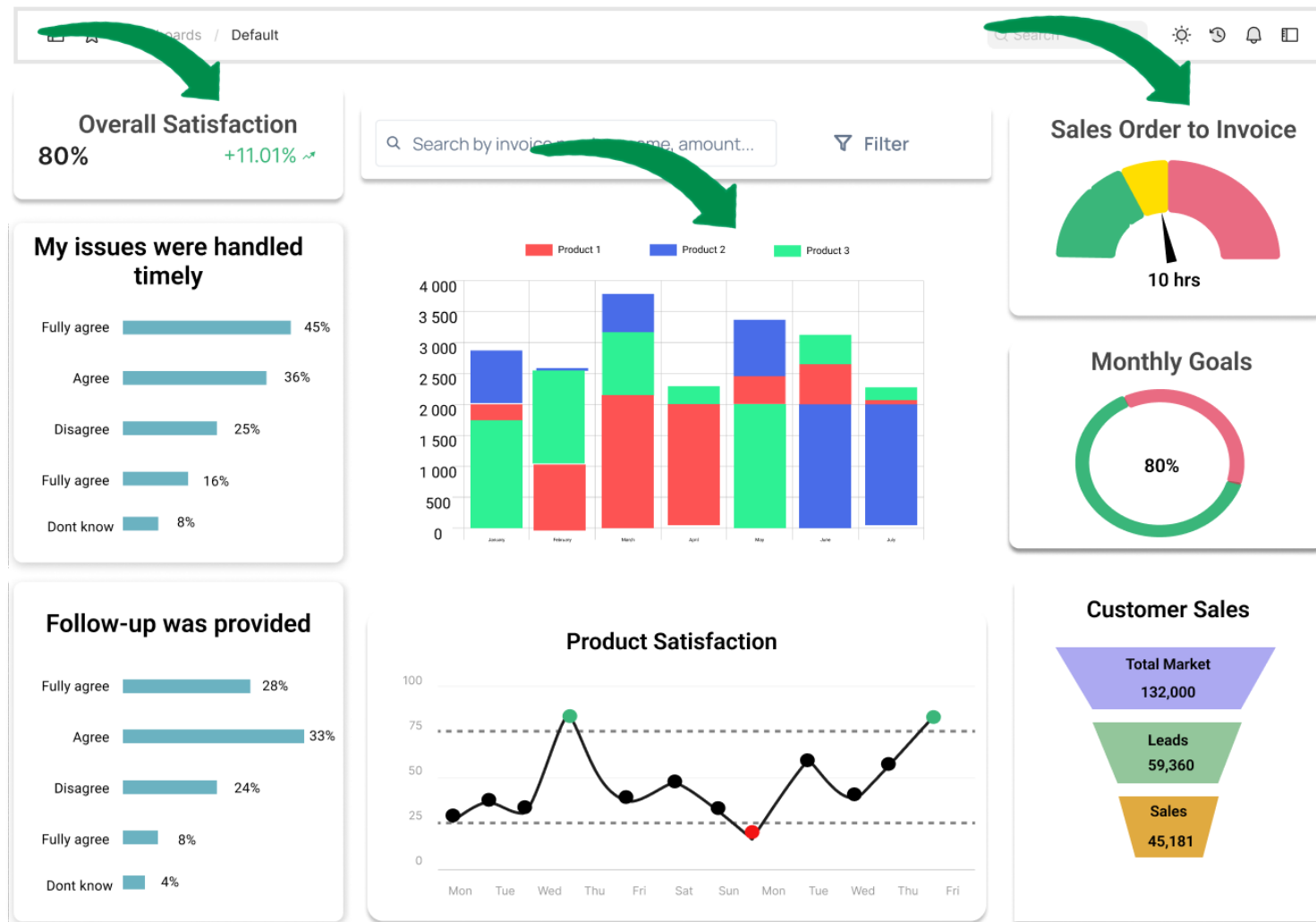


Status: Indicators

Target meter: goal setting and monitoring

¹ Zoho analytics. 2017. Analytical dashboard.

Analytical dashboard graphical components



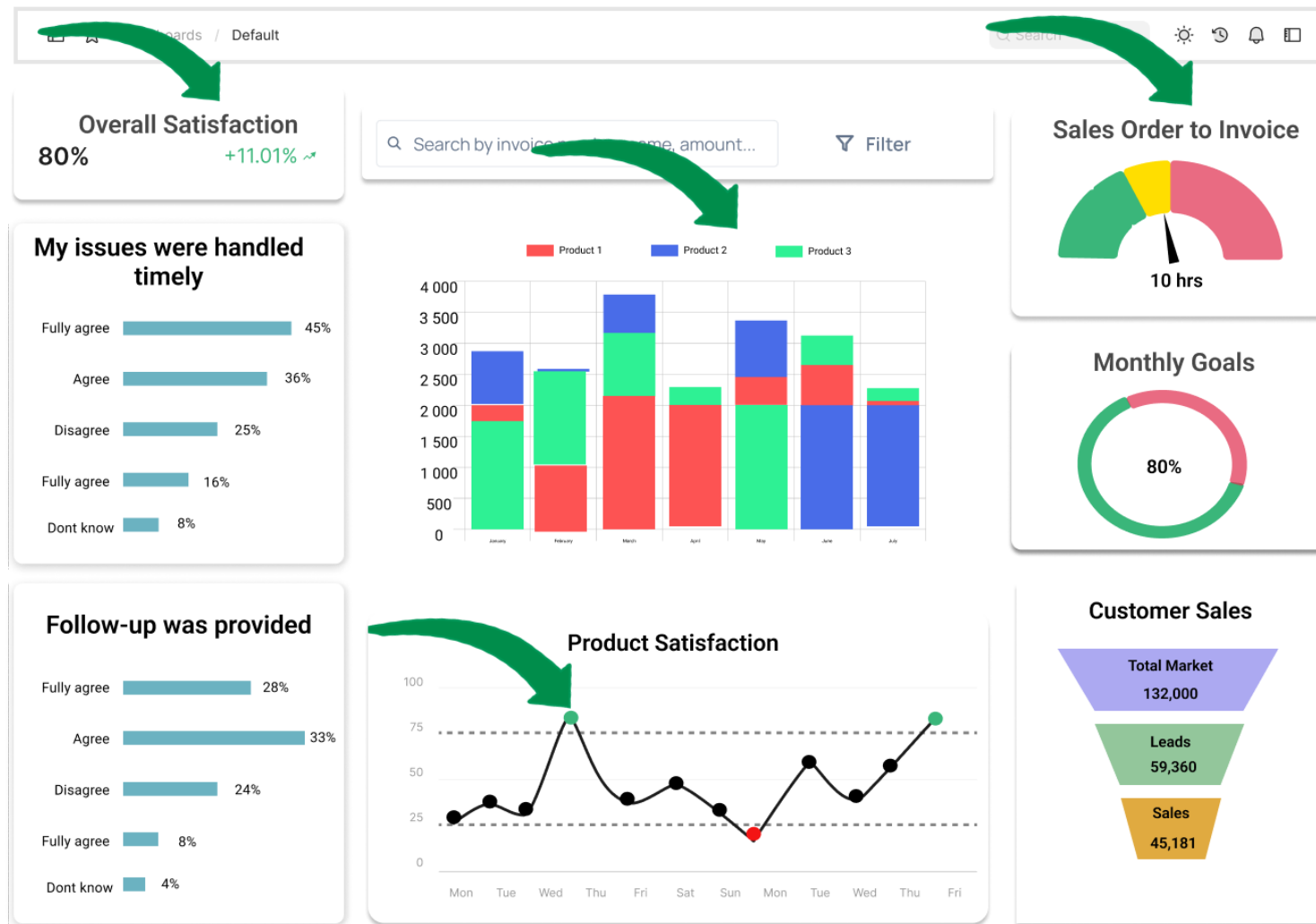
Status: Indicators

Target meter: goal setting and monitoring

Comparator: data overview and analogy

¹ Zoho analytics. 2017. Analytical dashboard.

Analytical dashboard graphical components



Status: Indicators

Target meter: goal setting and monitoring

Comparator: data overview and analogy

Anomaly detector: detection of business discrepancies

¹ Zoho analytics. 2017. Analytical dashboard.

Analytical dashboard graphical components



Status: Indicators

Target meter: goal setting and monitoring

Comparator: data overview and analogy

Anomaly detector: detection of business discrepancies

Funnel: visualization of the company's stages

¹ Zoho analytics. 2017. Analytical dashboard.

A special word about funnel charts

- Sequential stages of a business process (sales pipeline, marketing campaign outreach)
- Starts from a broad head and breaks down into smaller segments

Marketing campaign

Stage	Users	%
Sent	5676	100
Viewed	3872	68
Clicked	1668	29
Add to cart	610	11
Purchased	565	10

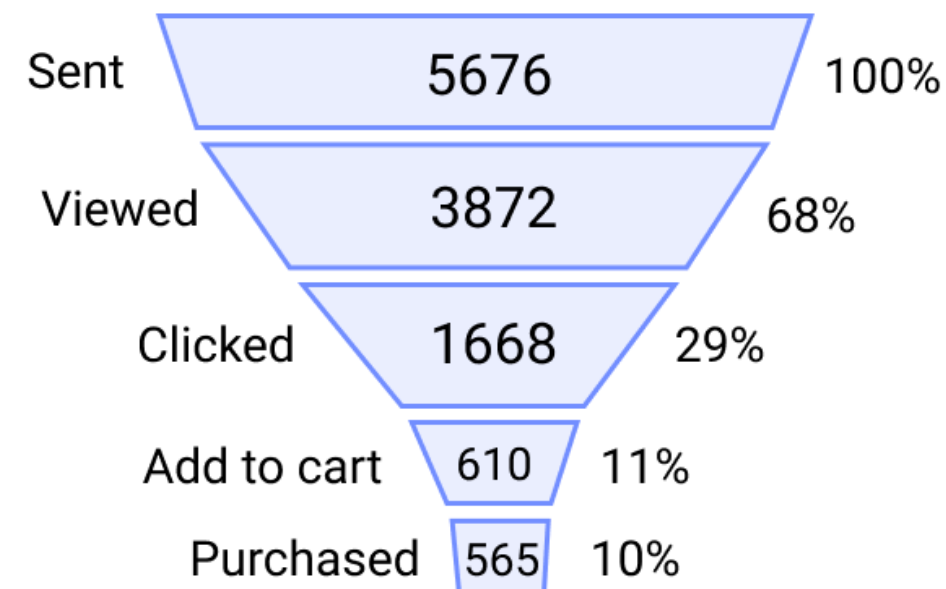
A special word about funnel charts

- Sequential stages of a business process (sales pipeline, marketing campaign outreach)
- Starts from a broad head and breaks down into smaller segments

Marketing campaign

Stage	Users	%
Sent	5676	100
Viewed	3872	68
Clicked	1668	29
Add to cart	610	11
Purchased	565	10

Funnel chart

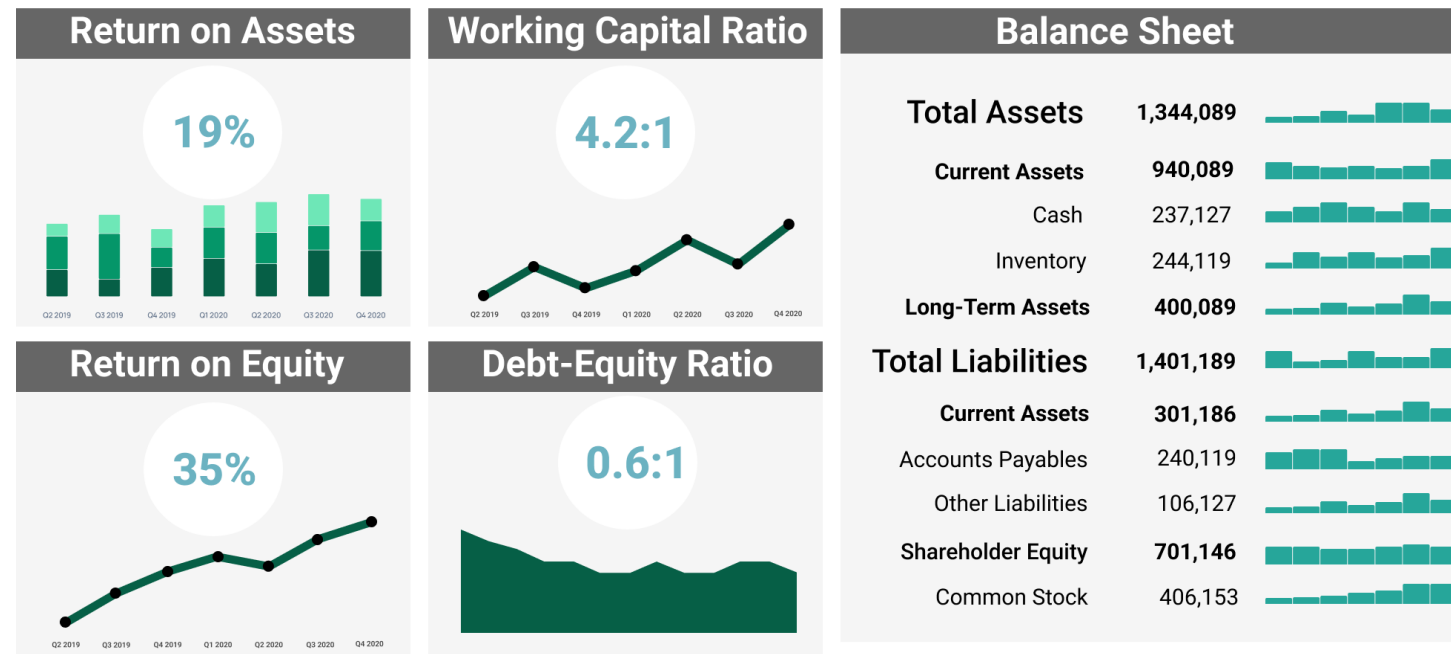


- Used as a high-level discovery visualization

Financial analytical dashboard

Objective: improving investment opportunities

Question: how efficiently the company's capital is spent



Task: set specific operational activities to improve future spending

Data: financial KPIs

- Efficiency ratios, working capital, balance sheet, income statement, cash flow

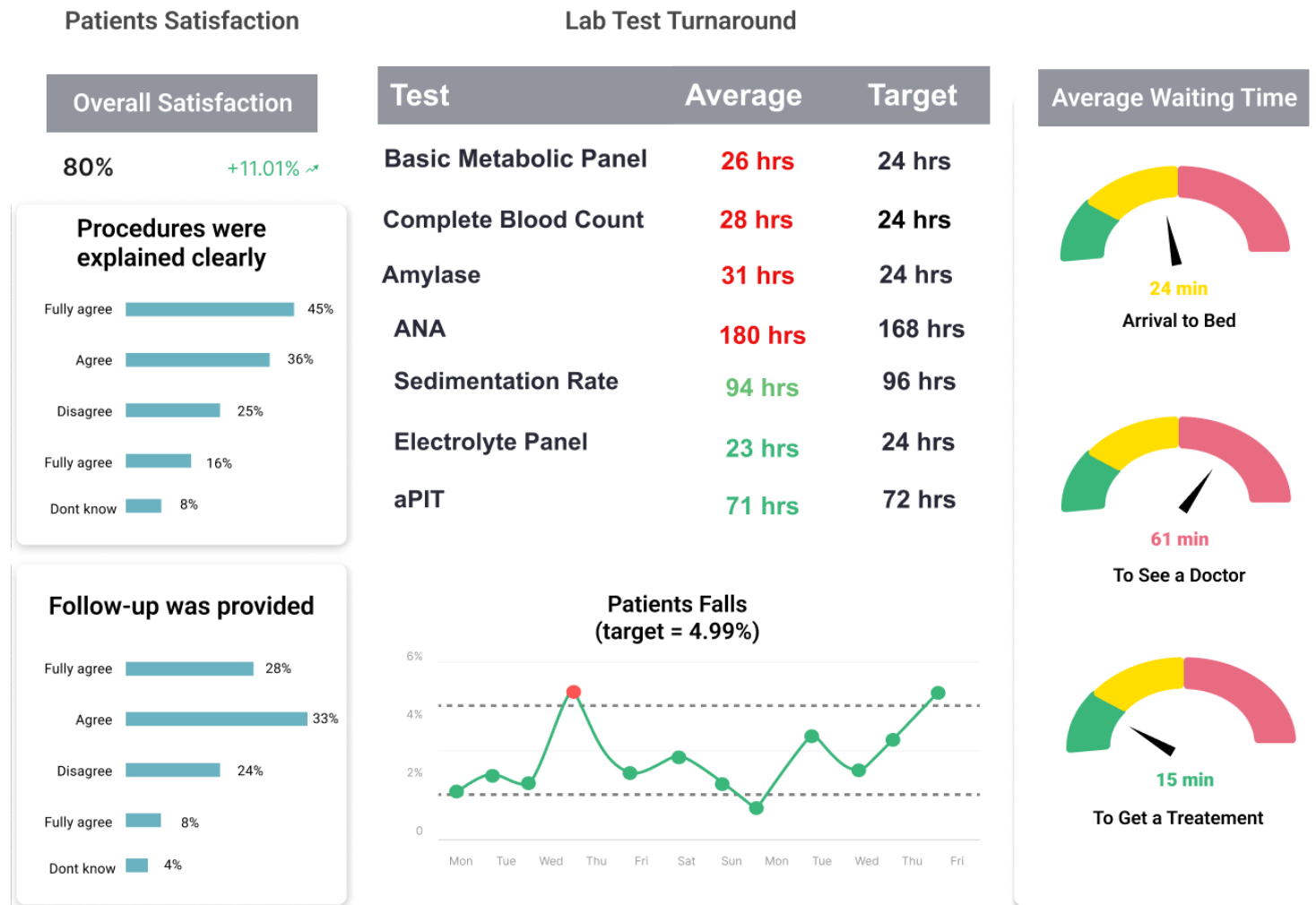
Data visualization: indicator, trend, comparator, anomaly detector

¹ Calzon, Bernardita. (2021). Dashboarding. Make Sure You Know The Difference.

Healthcare analytical dashboard

Objective: patients' care improvement

Question: how satisfied are the patients with the facility and the service



Task: provides set specific operational activities to improve service and facility

Data: healthcare KPIs

- Satisfaction, waiting times, and safety

Data visualization: indicator, target meters, trend, anomaly detector, color-coded table with current versus target turnaround (an alternative to bullet chart)

¹ Calzon, Bernardita. (2021). Dashboarding. Make Sure You Know The Difference.

Let's practice!
DASHBOARD DESIGN CONCEPTS

Strategic dashboard

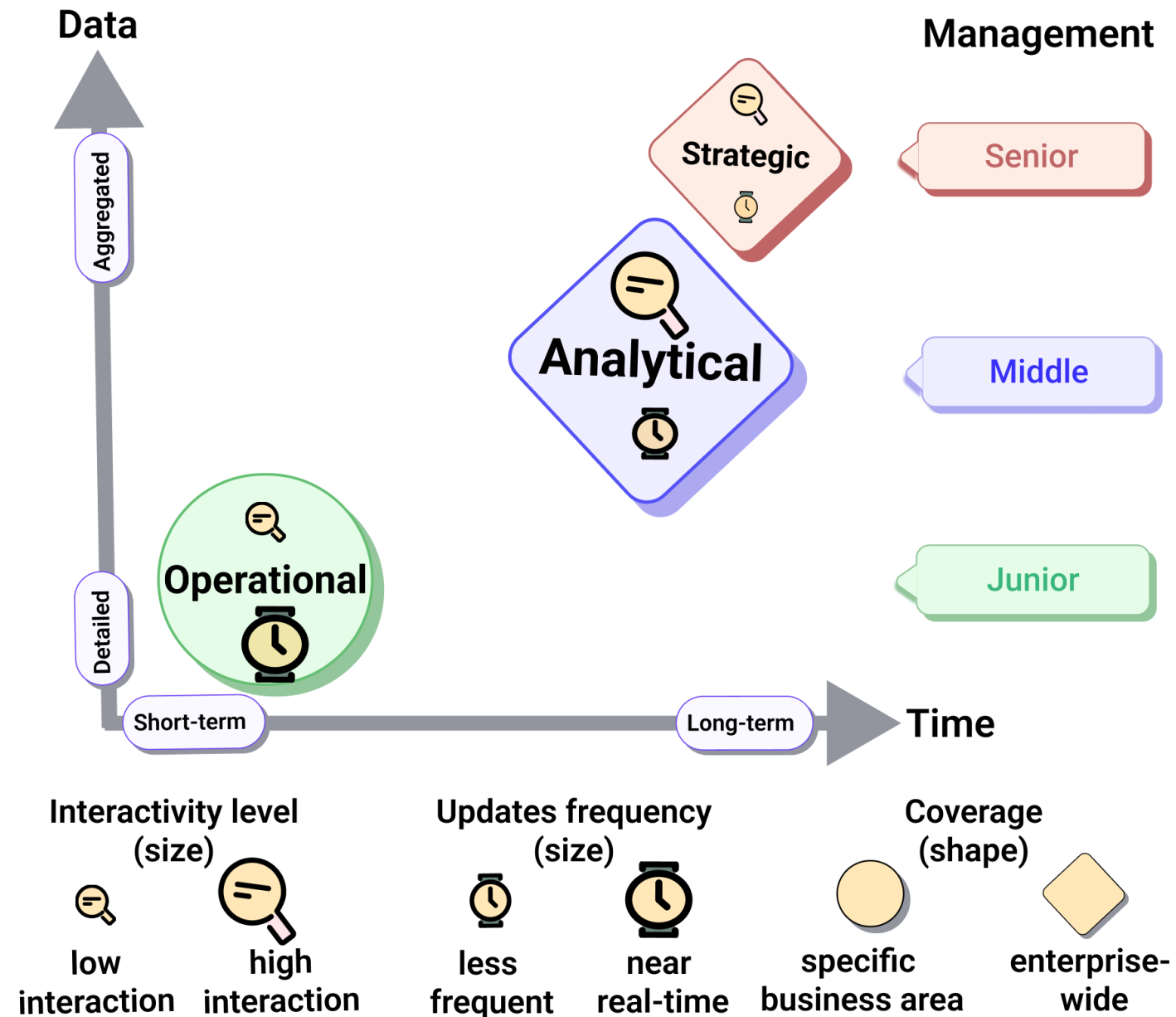
DASHBOARD DESIGN CONCEPTS



Olga Scrivner, PhD

President, Scrivner Solutions Inc

The forward-thinking dashboard



Purpose

- Monitor long-term company strategies

Data Type

- Complex, aggregated enterprise-wide data, infrequent updates

Interactivity:

- Low interaction, often static snapshots

Stakeholders

- Executives, senior-level management

Strategic dashboard benefits

Planning

- Planning and decision-making for a specific course of actions
 - Where did we fall short of our targets?
 - Which of the objectives have we completed?
 - Which objectives are no longer needed?

¹ Few, Stephen. (2006). Information dashboard design.

Strategic dashboard benefits

Planning

Comparison

- Planning and decision-making for specific course of actions
- Comparing current performance to target values for various success indicators

¹ Few, Stephen. (2006). Information dashboard design.

Strategic dashboard benefits

Planning

Comparison

Evaluation

- Planning and decision-making for specific course of actions
- Comparing current performance to target values for various success indicators
- Evaluating strategies (outdated and out-of-sync)
 - Is our strategic area still relevant?
 - Which of our objectives have we completed?

¹ Few, Stephen. (2006). Information dashboard design.

Strategic dashboard benefits

Planning

Comparison

Evaluation

Summary

- Planning and decision-making for specific course of actions
- Comparing current performance to target values for various success indicators
- Evaluating strategies (outdated and out-of-sync)
- Aggregating measures into high-level success indicators

¹ Few, Stephen. (2006). Information dashboard design.

Types of the strategic dashboard

Marketing: marketing KPIs

- Chief Marketing Officer

Executive: overall success KPIs

- Chief Administrative Officer

Financial: financial KPIs

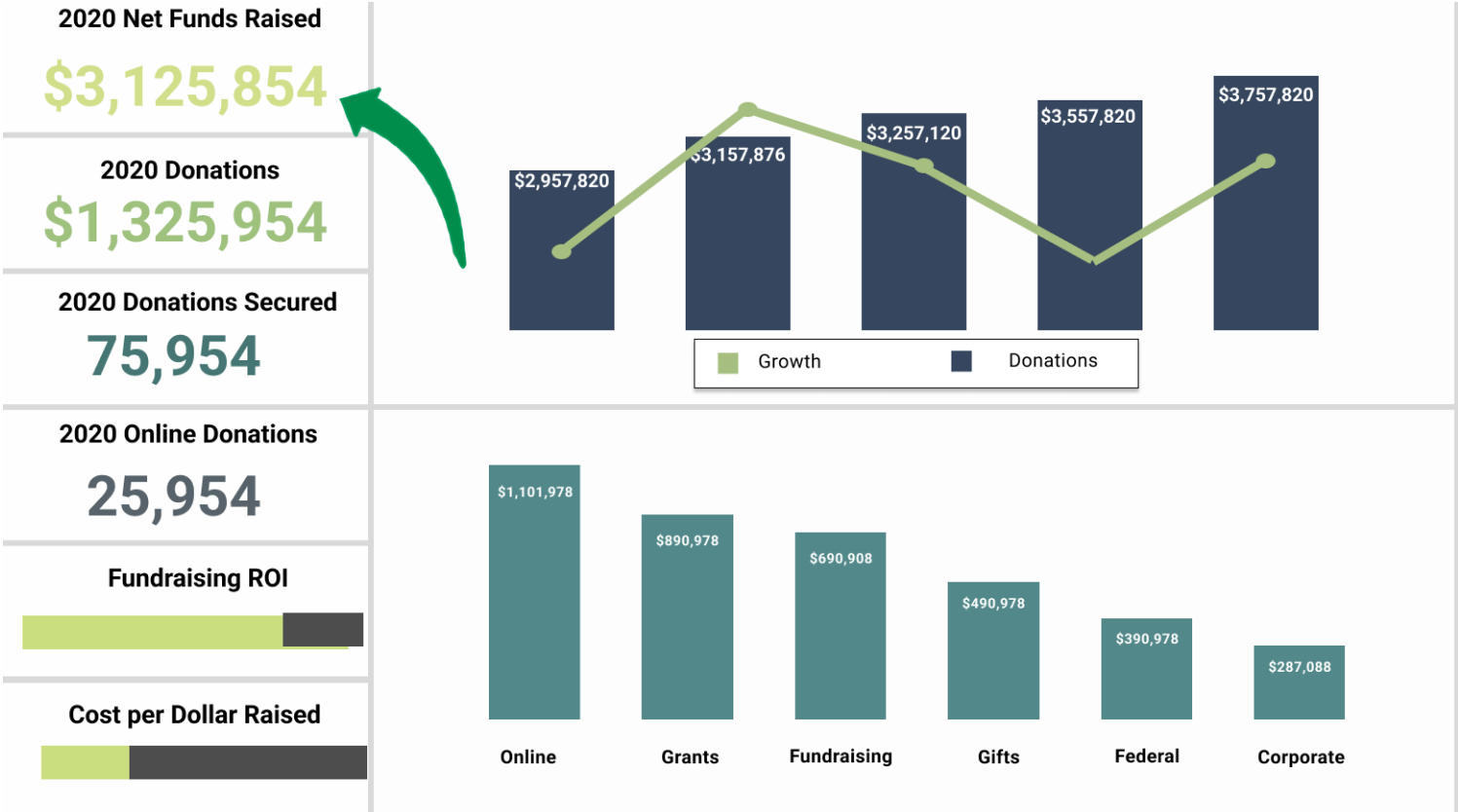
- Chief Financial Officer



Strategic dashboard graphical components



Strategic dashboard graphical components



Indicator: strategic success KPIs

Strategic dashboard graphical components



Indicator: single values of importance

Bar chart: specific values across different categories

Strategic dashboard graphical components



Indicator: single values of importance

Bar chart: specific values across different categories

Combo chart: two charts combined for a quick comparison

Strategic dashboard graphical components



Indicator: single values of importance

Bar chart: specific values across different categories

Combo chart: two charts combined for a quick comparison

Bullet chart: performance against a target

A special word about bullet graphs



¹ Allio, Michael. (2012). Strategic dashboards.

A special word about bullet graphs



- Best used for making comparisons between actual and target values

¹ Allio, Michael. (2012). Strategic dashboards.

A special word about bullet graphs



- Best used for making comparisons between actual and target values
- Not suitable for analyzing change-over-time or showing a distribution

¹ Allio, Michael. (2012). Strategic dashboards.

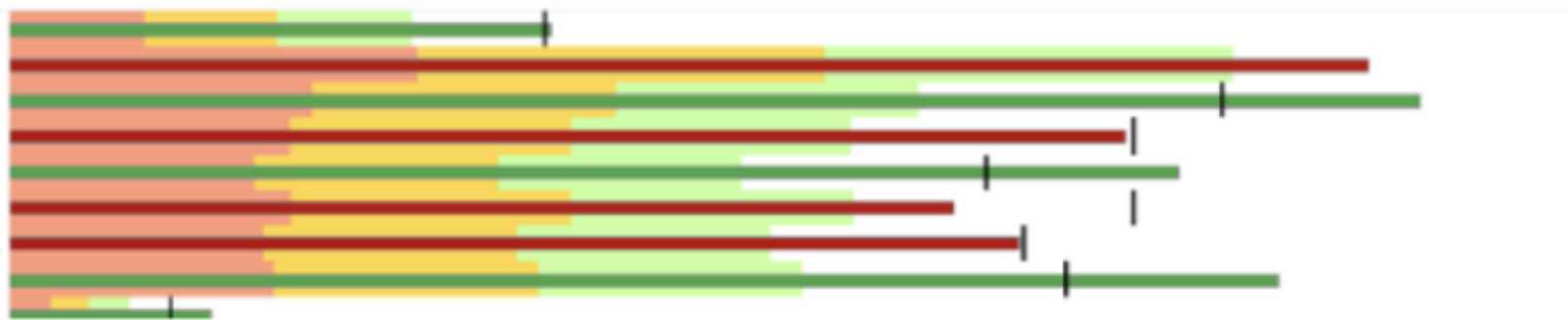
A special word about bullet graphs



- Best used for making comparisons between actual and target values
- Not suitable for analyzing change-over-time or showing a distribution
- Reference line is the progress toward a goal

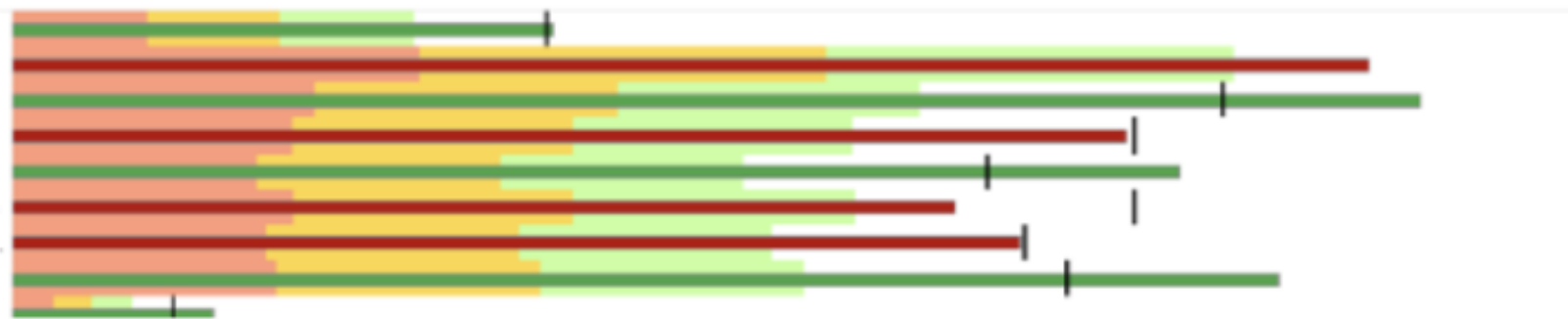
¹ Allio, Michael. (2012). Strategic dashboards.

A special word about bullet graphs



¹ Allio, Michael. (2012). Strategic dashboards.

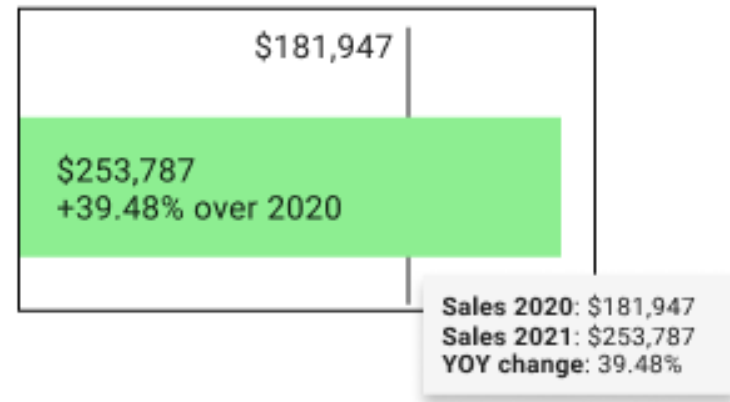
A special word about bullet graphs



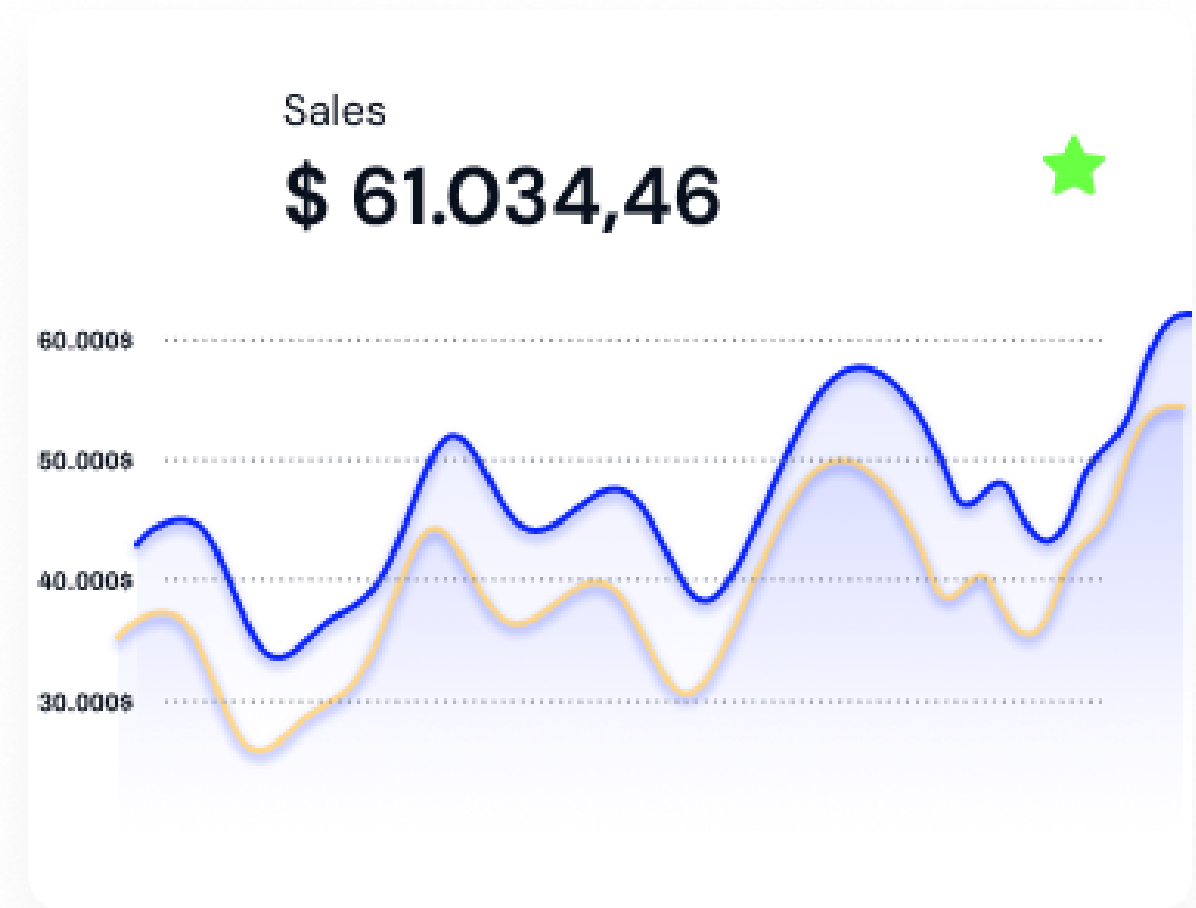
¹ Allio, Michael. (2012). Strategic dashboards.

A special word about period over period graph

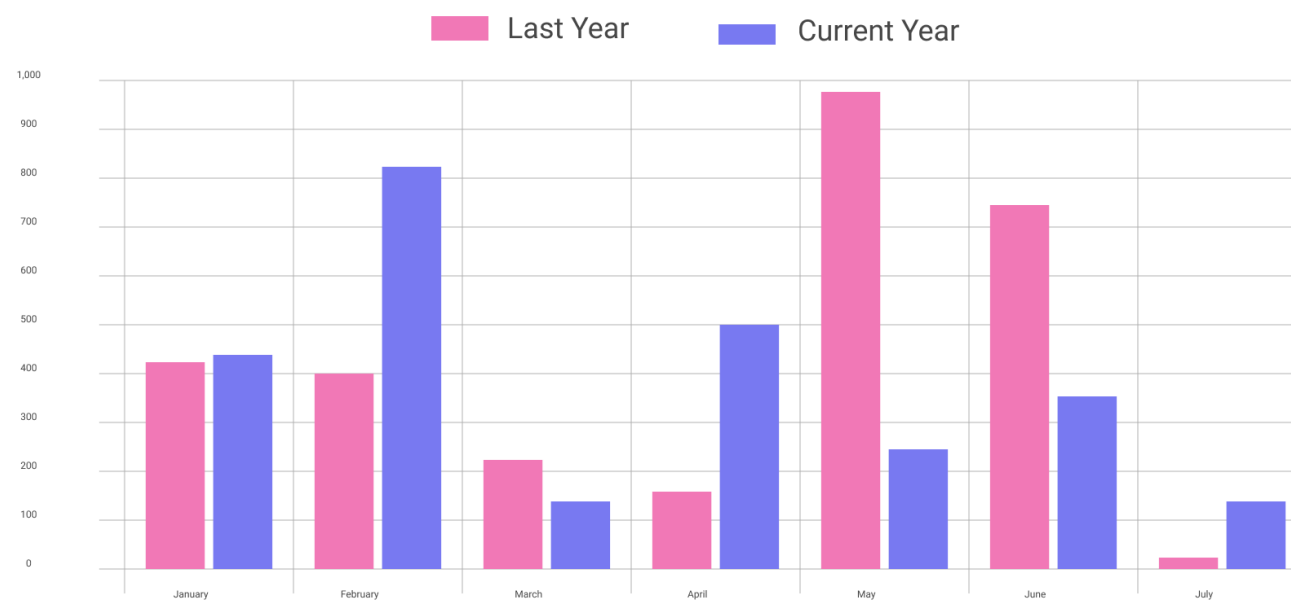
KPI indicator



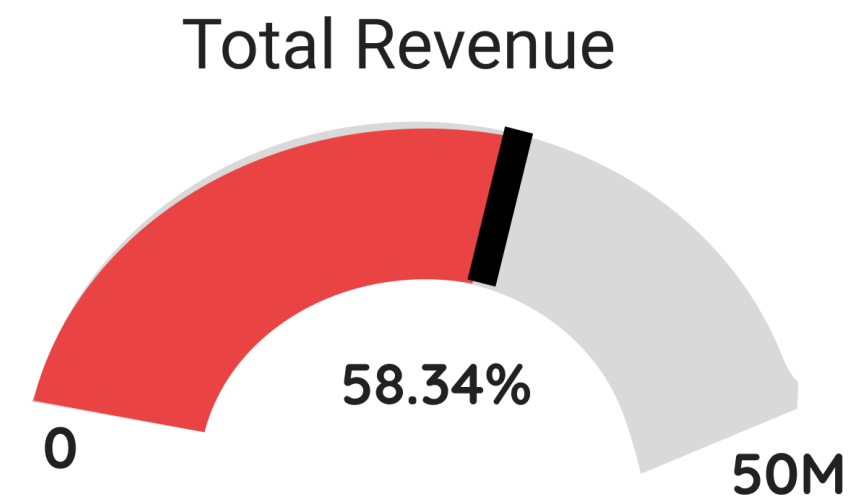
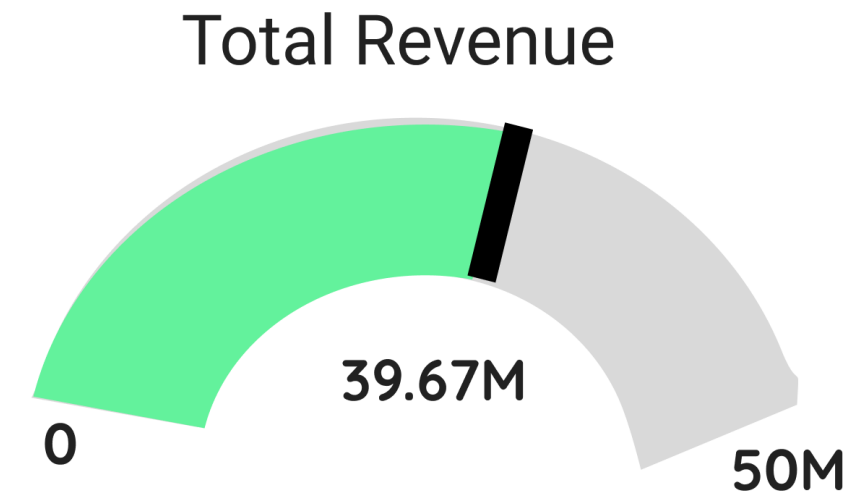
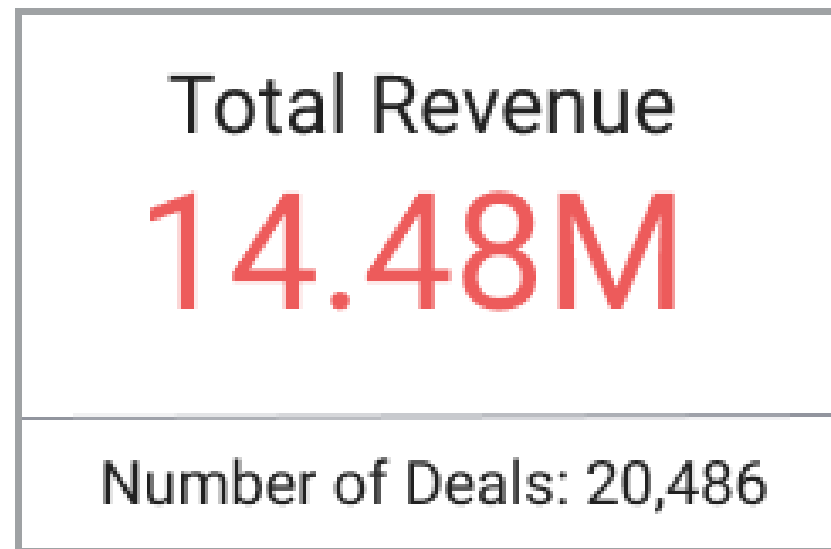
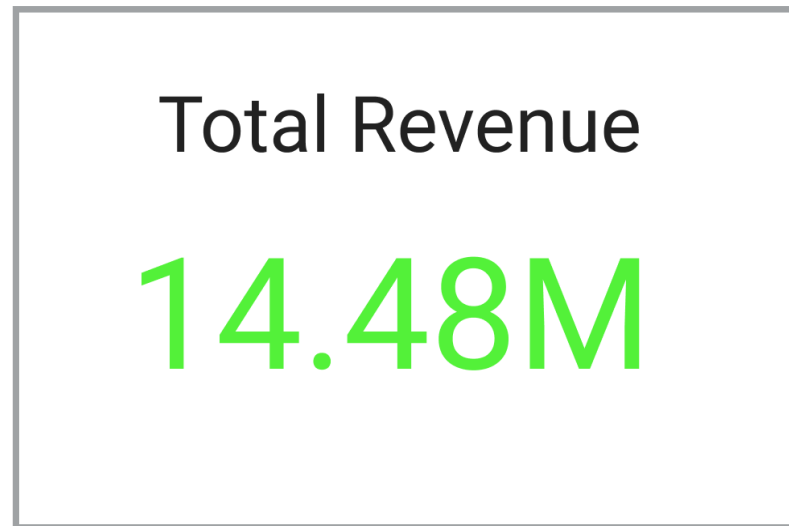
Line chart



Column chart



More about indicators and gauge charts



Let's practice!
DASHBOARD DESIGN CONCEPTS

Tactical dashboard

DASHBOARD DESIGN CONCEPTS



Olga Scrivner, PhD

President, Scrivner Solutions Inc

A focused insights dashboard

Purpose

- Analyzing a particular area of business to track the company's goal

Data Type

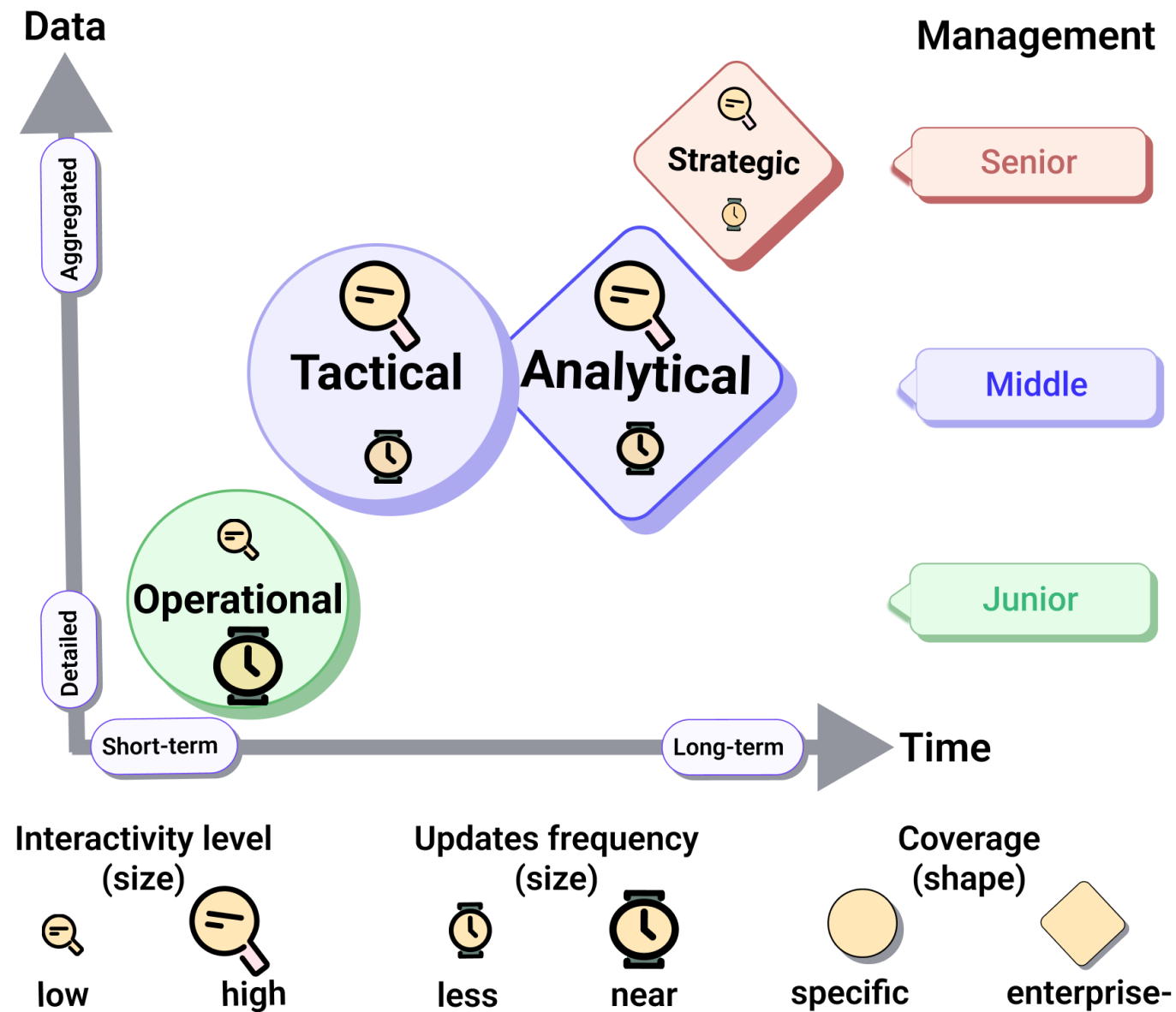
- Historical data, updated less frequently

Interactivity

- Filter, search, drill-down

Stakeholders

- Middle management



Tactical dashboard benefits

Detailed information

- Exploring in-depth various areas of focus using filters and segmentation

Tactical dashboard benefits

Detailed information

Tracking trends

- Exploring in-depth various areas of focus using filters and segmentation
- Using trend analysis to compare current performance with the past performance

Tactical dashboard benefits

Detailed information

Tracking trends

Monitoring performance

- Exploring in-depth various areas of focus using filters and segmentation
- Using trend analysis to compare current performance with the past performance
- Monitoring performance in relation to company's goals

Tactical dashboard benefits

Detailed information

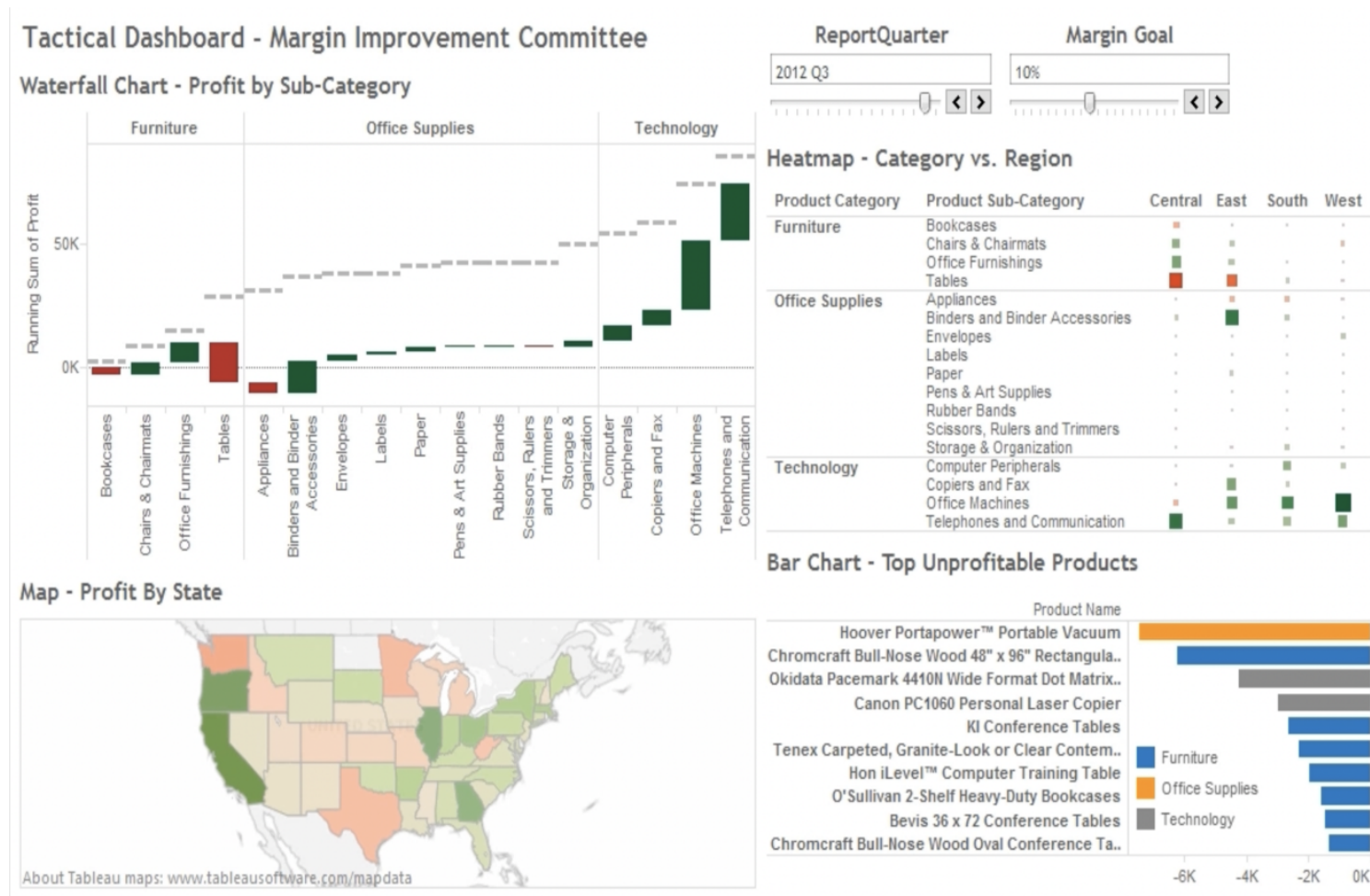
Tracking trends

Monitoring performance

Exposing issues

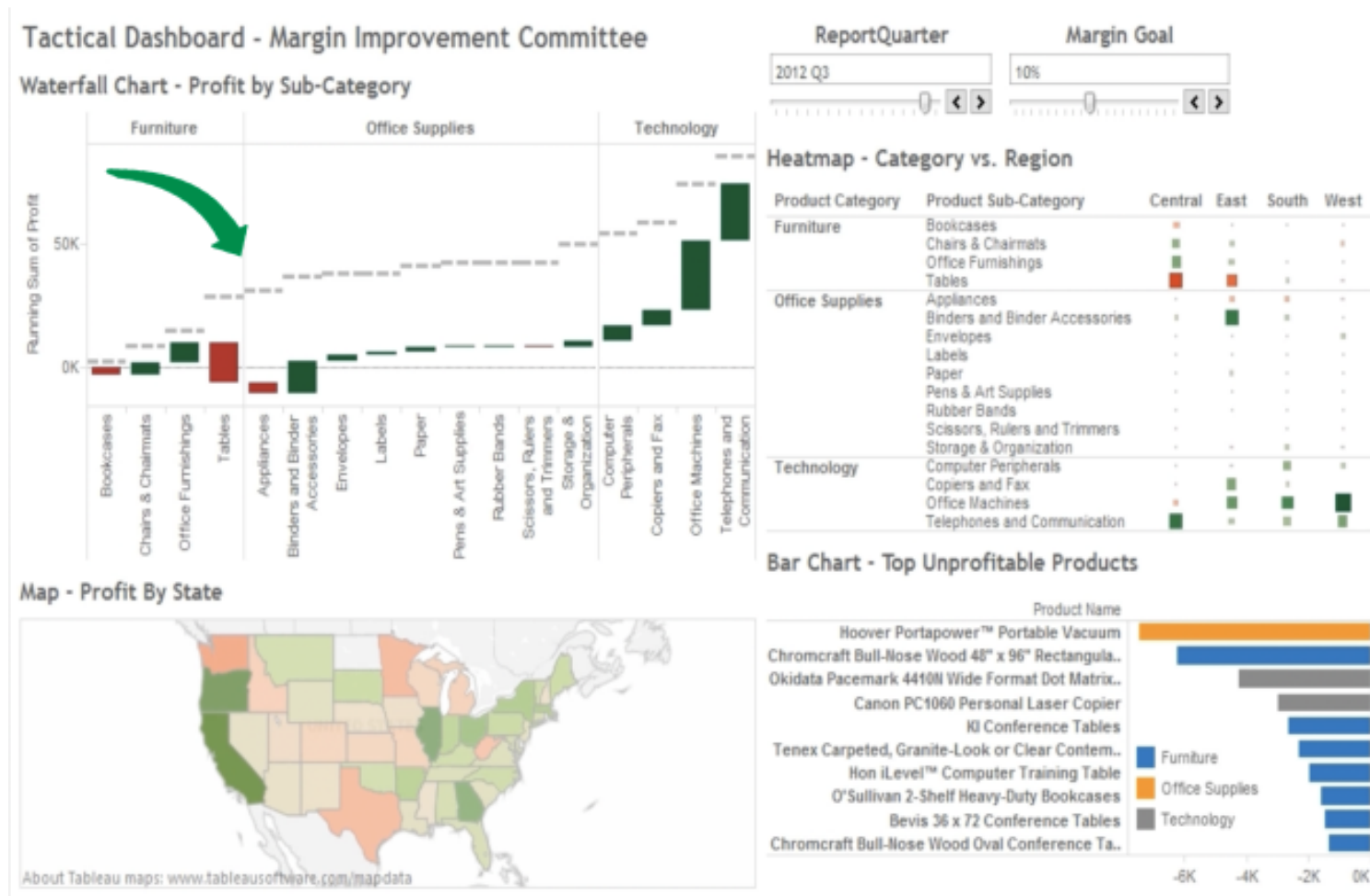
- Exploring in-depth various areas of focus using filters and segmentation
- Using trend analysis to compare current performance with the past performance
- Monitoring performance in relation to company's goals
- Setting early alarm for any anomalies within the current process

Tactical dashboard graphical components



Indicator: focused KPIs

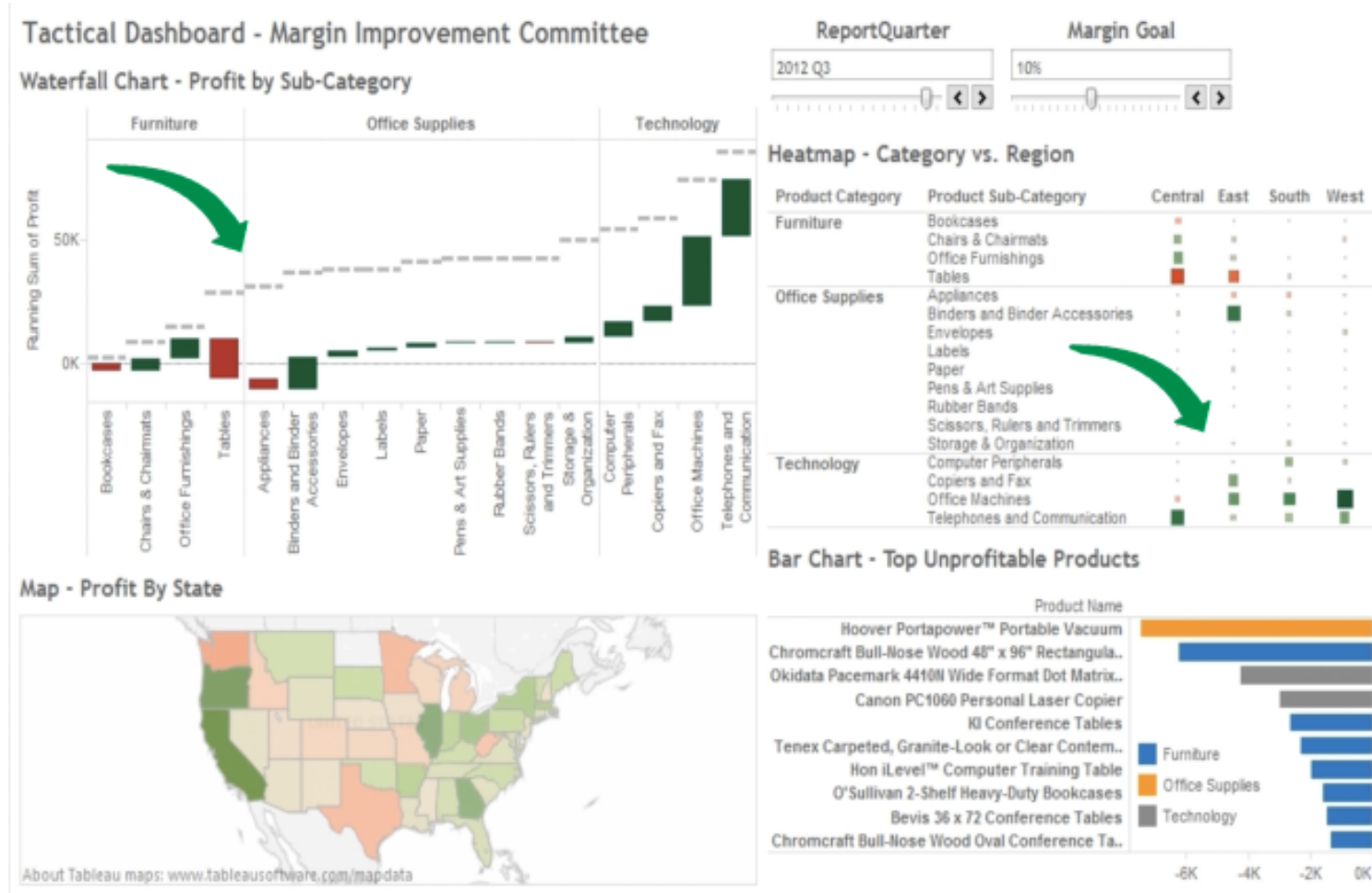
Tactical dashboard graphical components



Indicator: focused KPIs

Waterfall chart: compare parts to the whole (quarterly performance to the goal)

Tactical dashboard graphical components

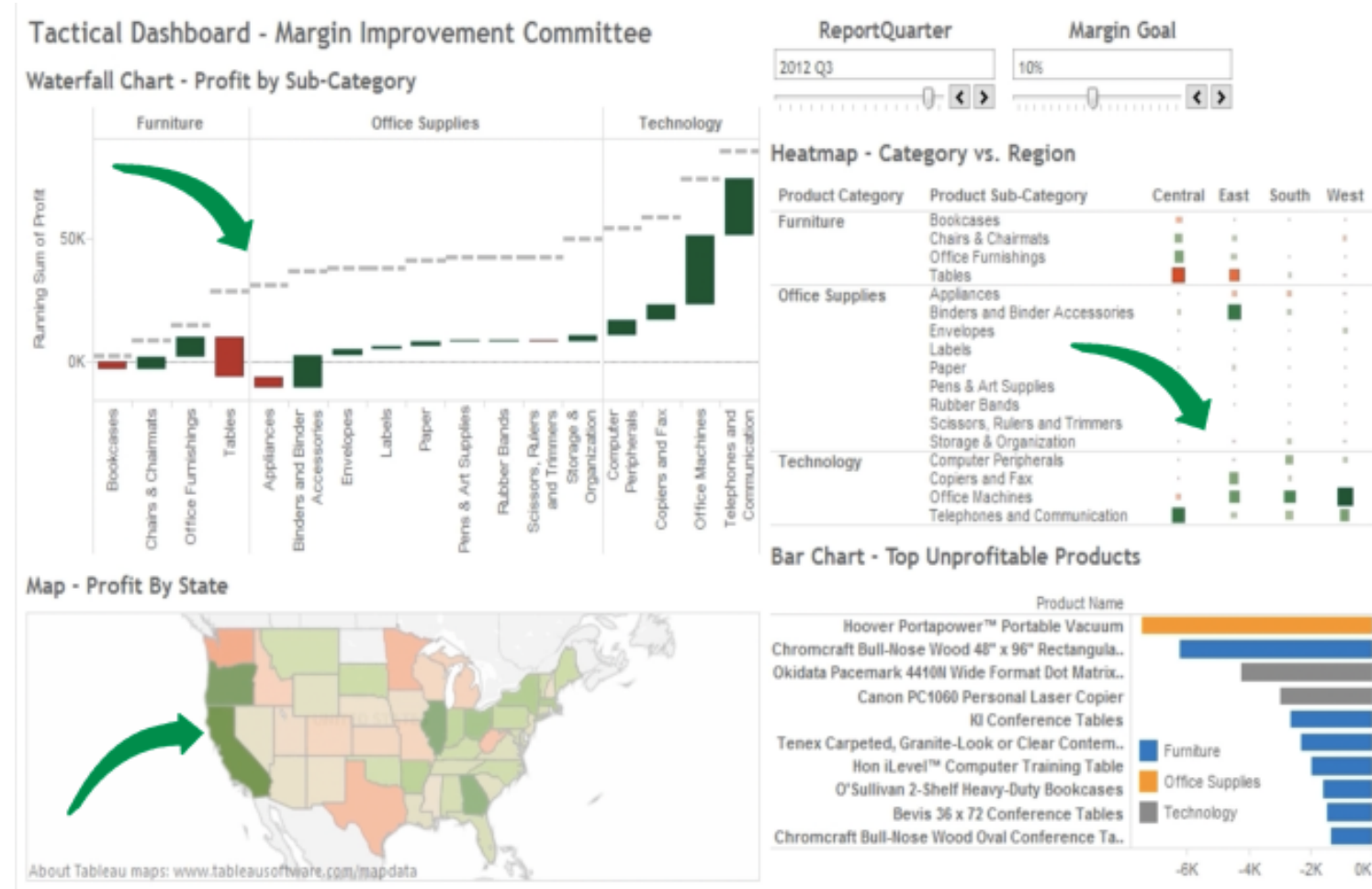


Indicator: focused KPIs

Waterfall chart: compare parts to the whole (quarterly performance to the goal)

Heatmap: the information is coded by color, intensity, size in a table-like format

Tactical dashboard graphical components



Indicator: focused KPIs

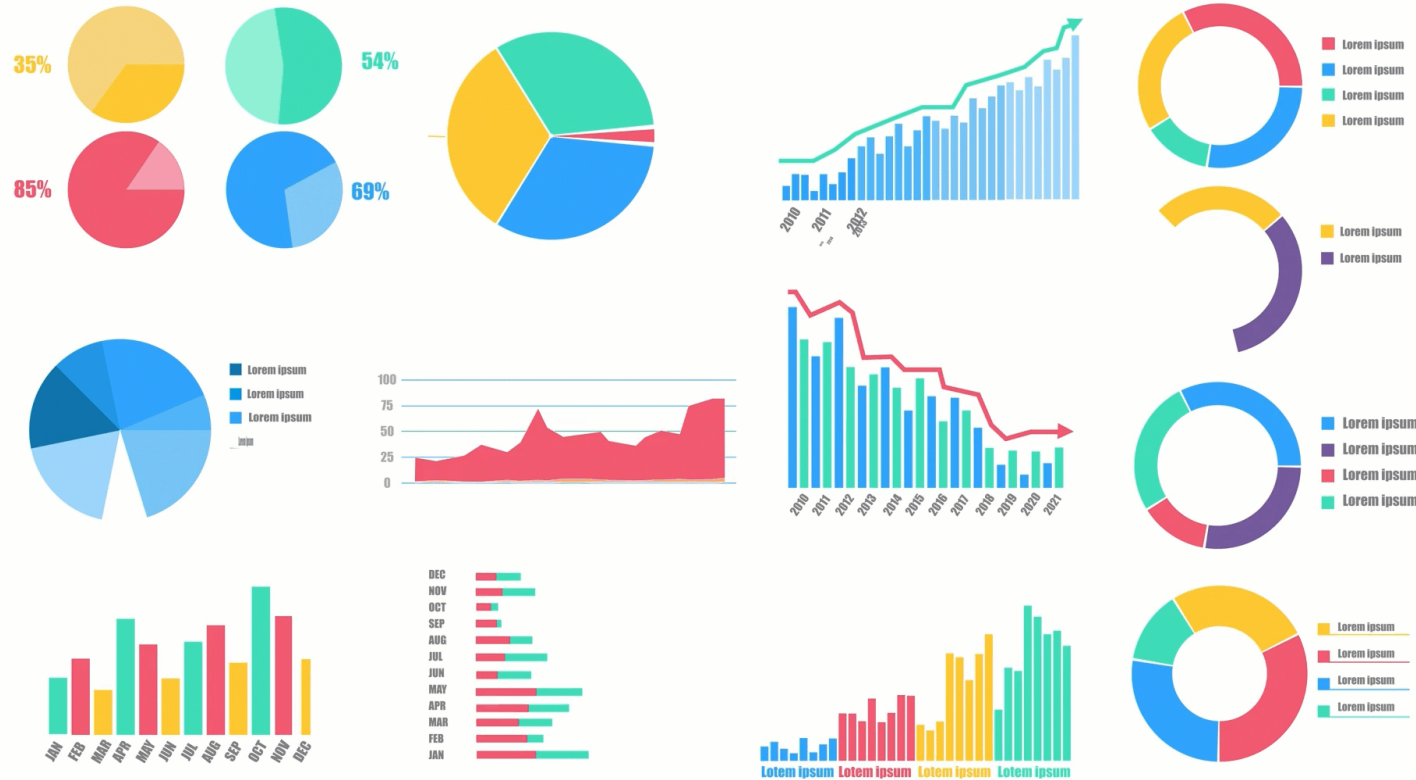
Waterfall chart: compare parts to the whole (quarterly performance to the goal)

Heatmap: the information is coded by color, intensity, size in a table-like format

Geographical map: information is presented by location

¹ Kau, Tony. (2014). Chapter 3: Tactical Dashboards

Tactical dashboard examples



Information technology dashboard: improving efficiency and optimizing projects

Energy dashboard: consumption and production costs

Human resource dashboard: reducing turnover, enhance employee satisfaction

Social media dashboard: keeping connection, increasing engagement, expanding the reach

Key Takeaways

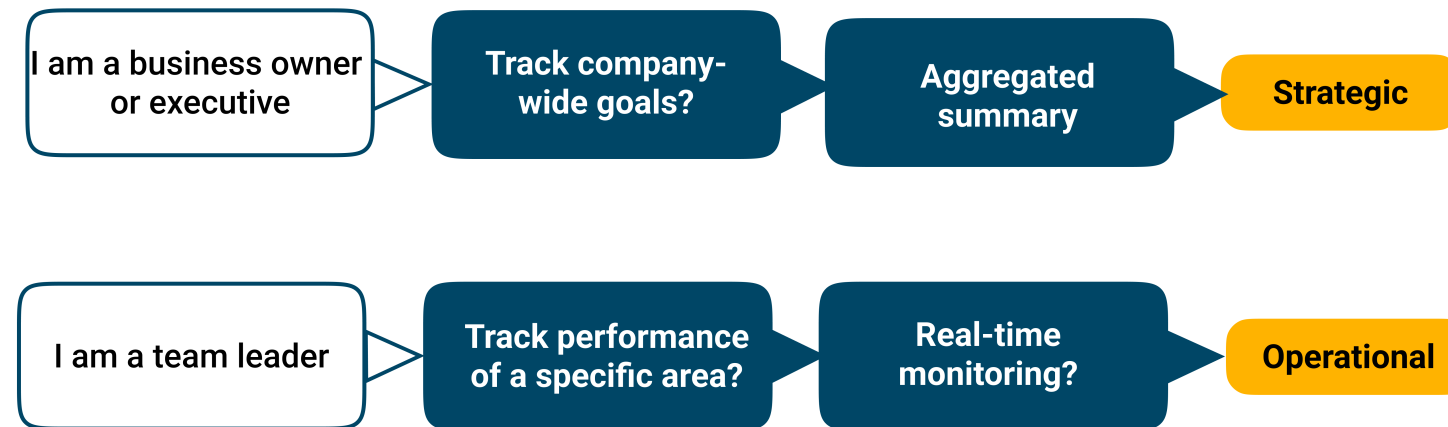
Each type: a different stakeholder and goal



Strategic - long-term strategies and high-level metrics for executives

Key Takeaways

Each type: a different stakeholder and goal

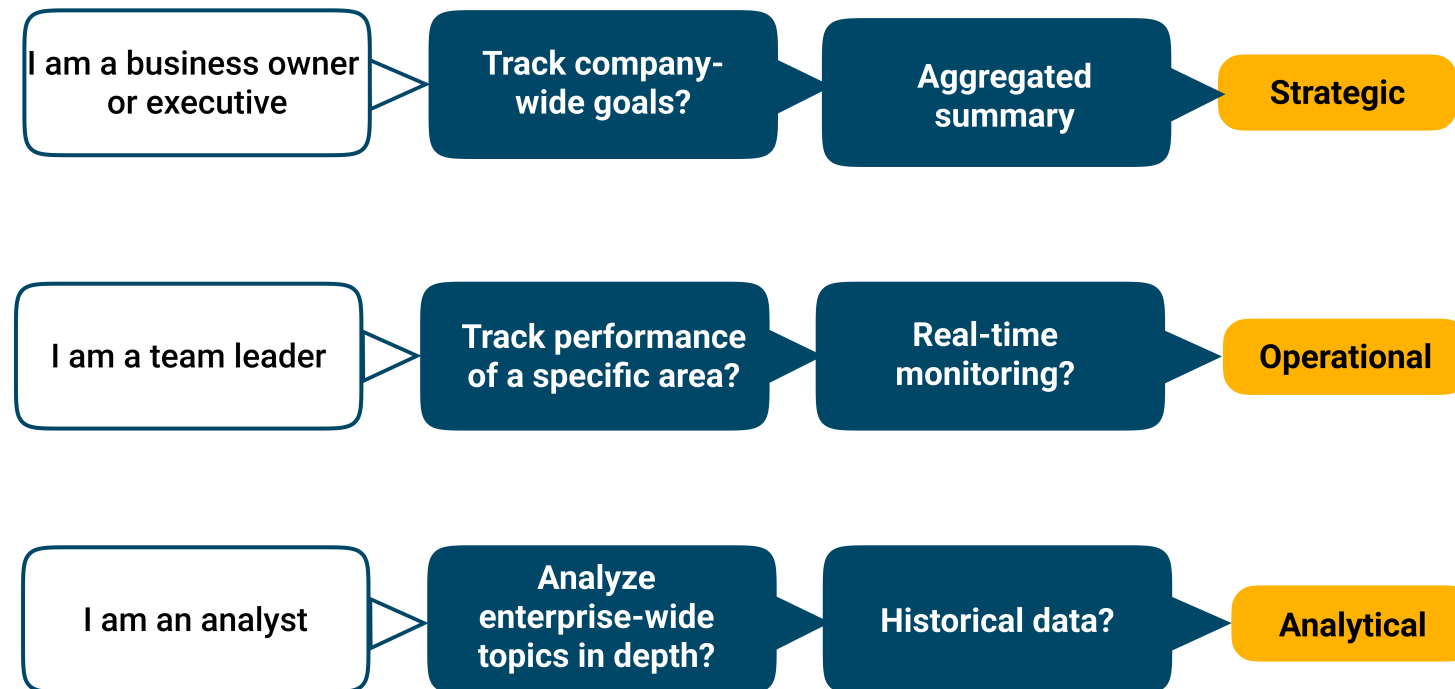


Strategic - long-term strategies and high-level metrics for executives

Operational - real-time and operational processes for junior management

Key Takeaways

Each type: a different stakeholder and goal



Strategic - long-term strategies and high-level metrics for executives

Operational - real-time and operational processes for junior management

Analytical - vast amounts of data for analysts to gain new insights

Key Takeaways

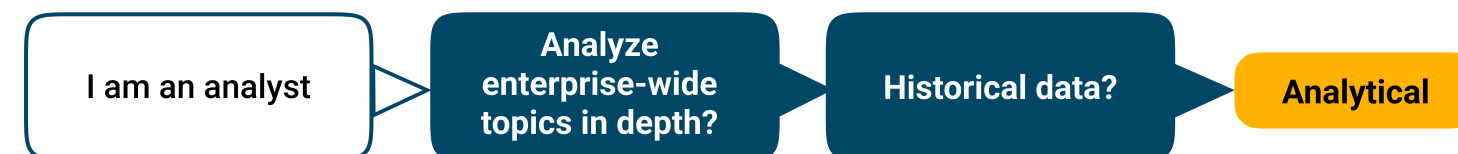
Each type: a different stakeholder and goal



Strategic - long-term strategies and high-level metrics for executives



Operational - real-time and operational processes for junior management



Analytical - vast amounts of data for analysts to gain new insights



Tactical - mid-management to track and analyze performance

Let's practice!
DASHBOARD DESIGN CONCEPTS